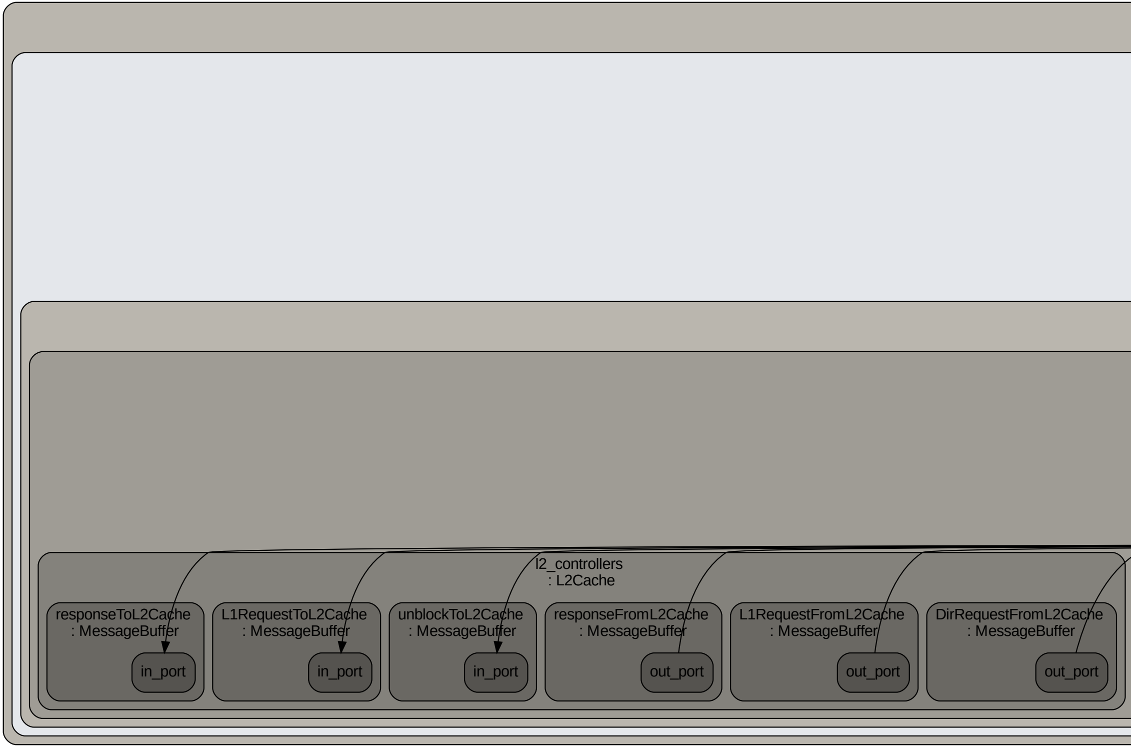
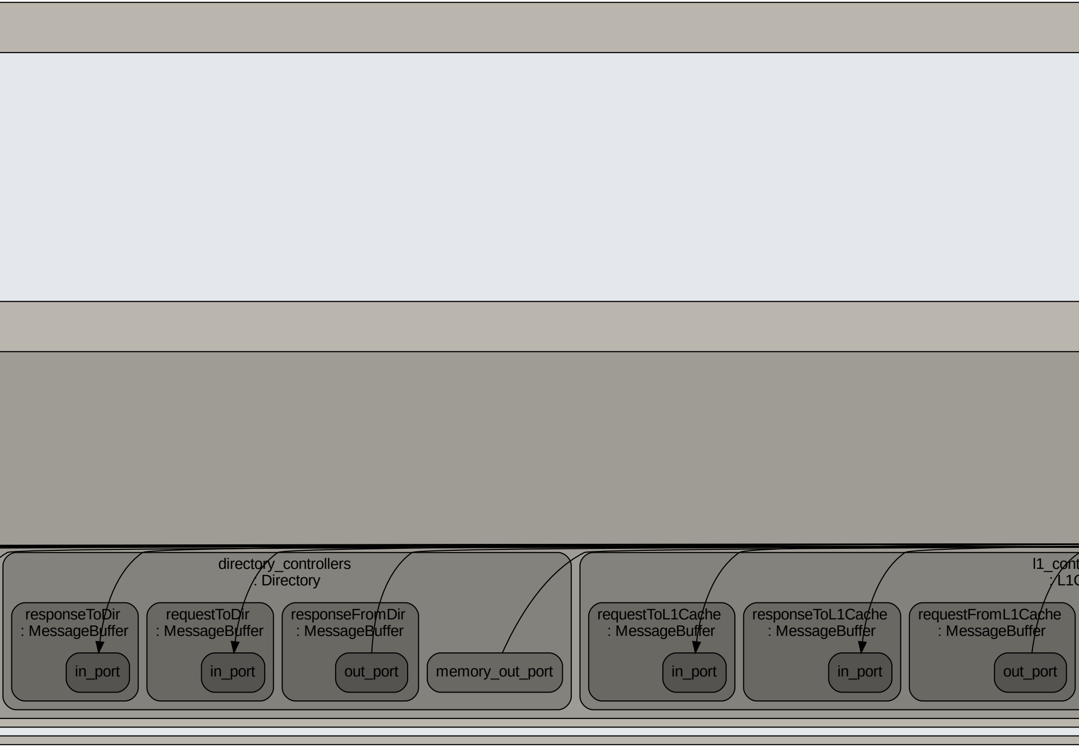
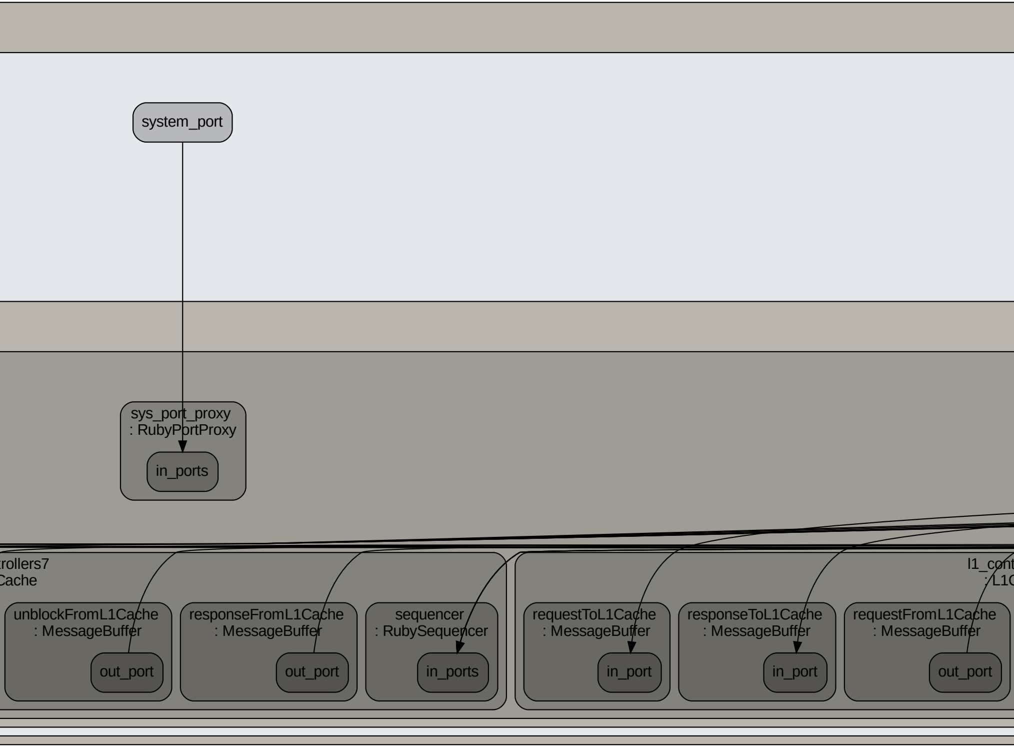
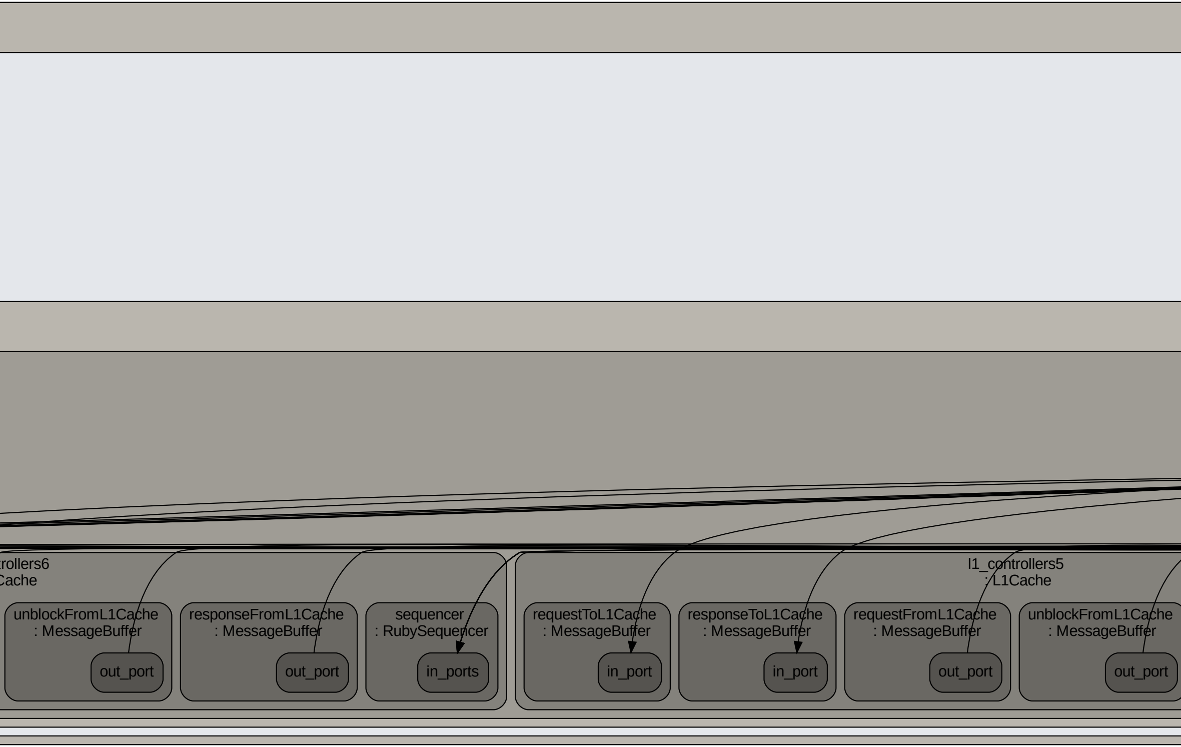
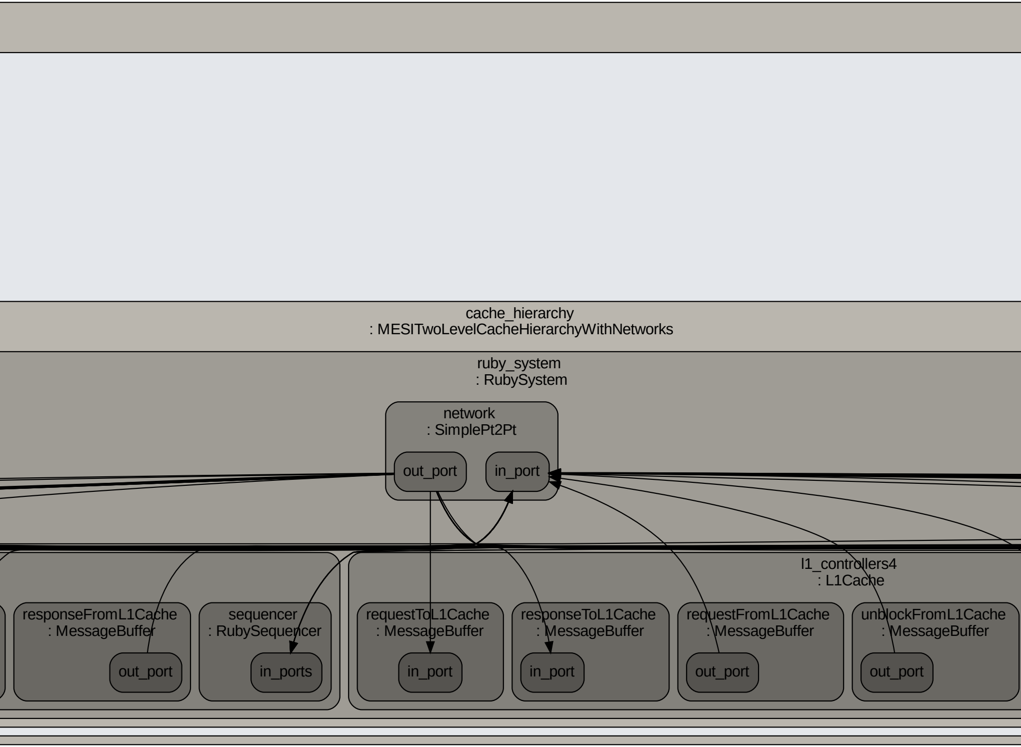
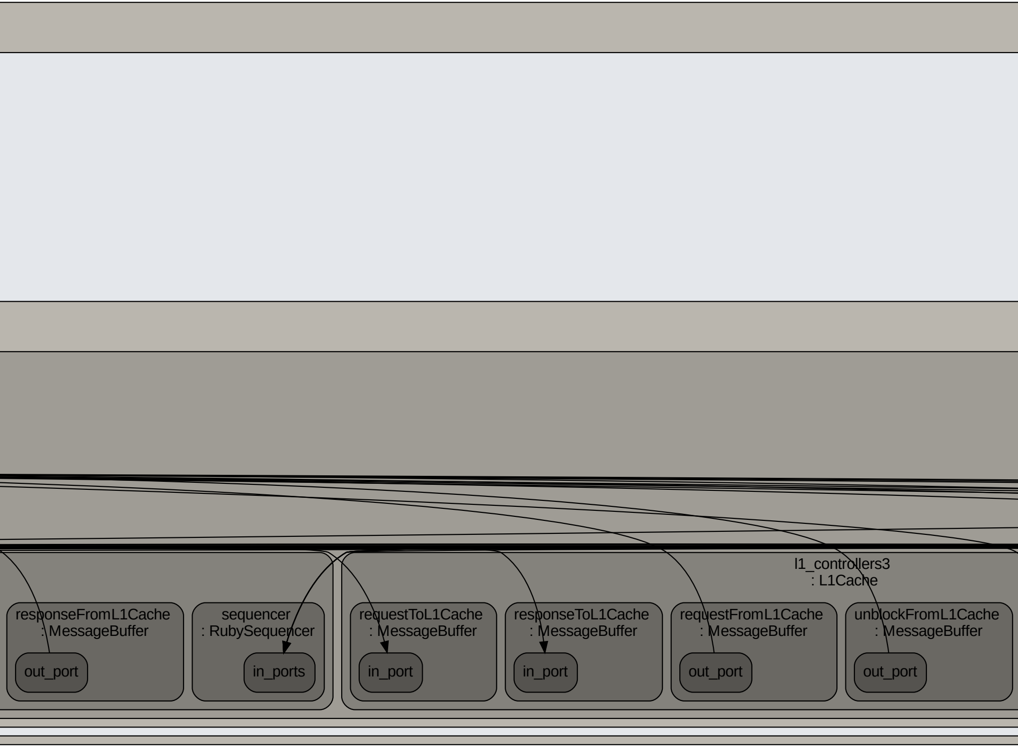

l2_controllers
: L2Cache
responseToL2Cache
: MessageBuffer
in_port
L1RequestToL2Cache
: MessageBuffer
in_port
unblockToL2Cache
: MessageBuffer
in_port
responseFromL2Cache
: MessageBuffer
out_port
L1RequestFromL2Cache
: MessageBuffer
out_port
DirRequestFromL2Cache
: MessageBuffer
out_port

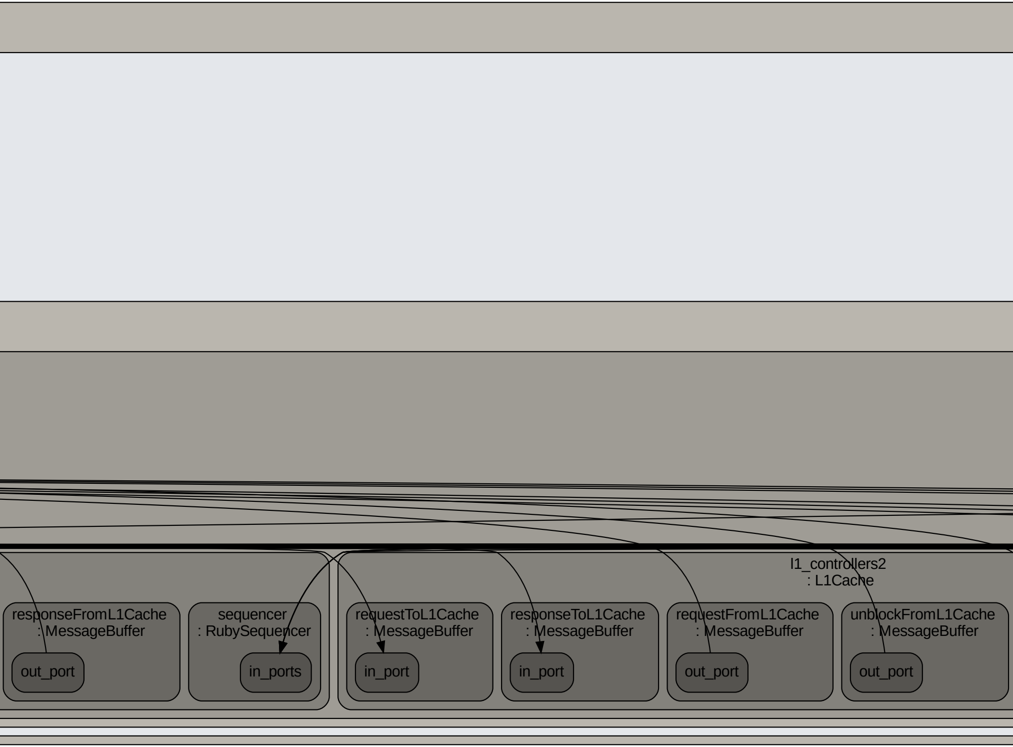

directory_controllers
: Directory
responseToDir
: MessageBuffer
in_port
requestToDir
: MessageBuffer
in_port
responseFromDir
: MessageBuffer
out_port
memory_out_port
requestToL1Cache
: MessageBuffer
in_port
responseToL1Cache
: MessageBuffer
in_port
requestFromL1Cache
: MessageBuffer
out_port
l1_cont
: L1C

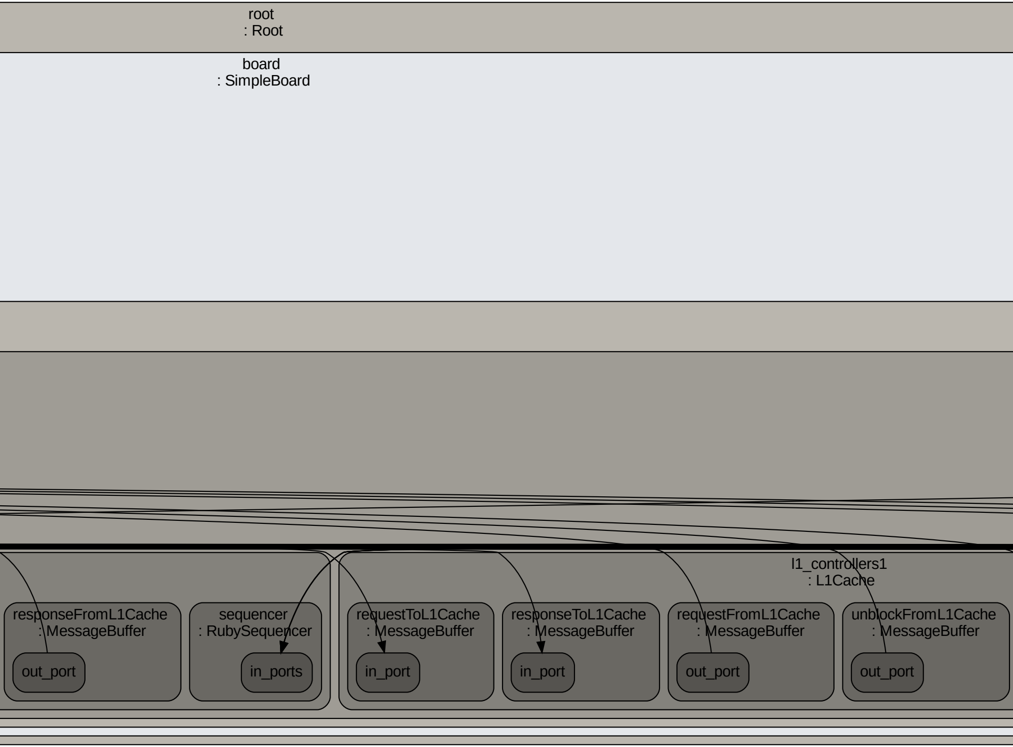

system_port
sys_port_proxy
: RubyPortProxy
in_ports
trollers7
Cache
unblockFromL1Cache
: MessageBuffer
out_port
responseFromL1Cache
: MessageBuffer
out_port
sequencer
: RubySequencer
in_ports
l1_cont
: L1C
requestToL1Cache
: MessageBuffer
in_port
responseToL1Cache
: MessageBuffer
in_port
requestFromL1Cache
: MessageBuffer
out_port

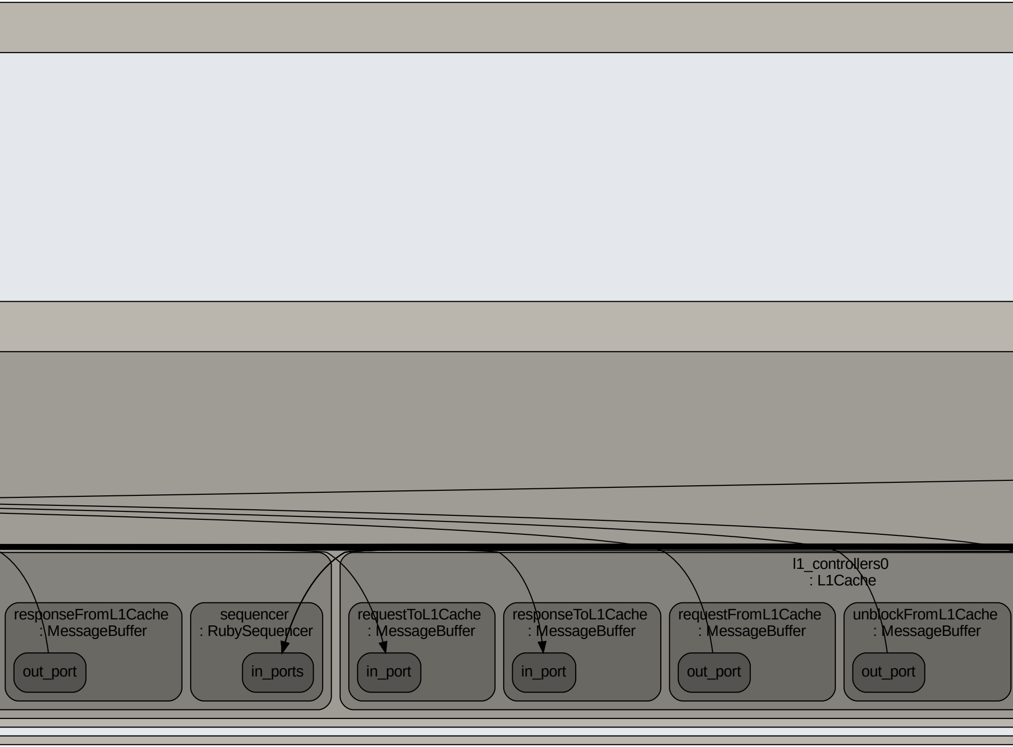

trollers6
Cache
unblockFromL1Cache
: MessageBuffer
out_port
responseFromL1Cache
: MessageBuffer
out_port
sequencer
: RubySequencer
in_ports
l1_controllers5
: L1Cache
requestToL1Cache
: MessageBuffer
in_port
responseToL1Cache
: MessageBuffer
in_port
requestFromL1Cache
: MessageBuffer
out_port
unblockFromL1Cache
: MessageBuffer
out_port

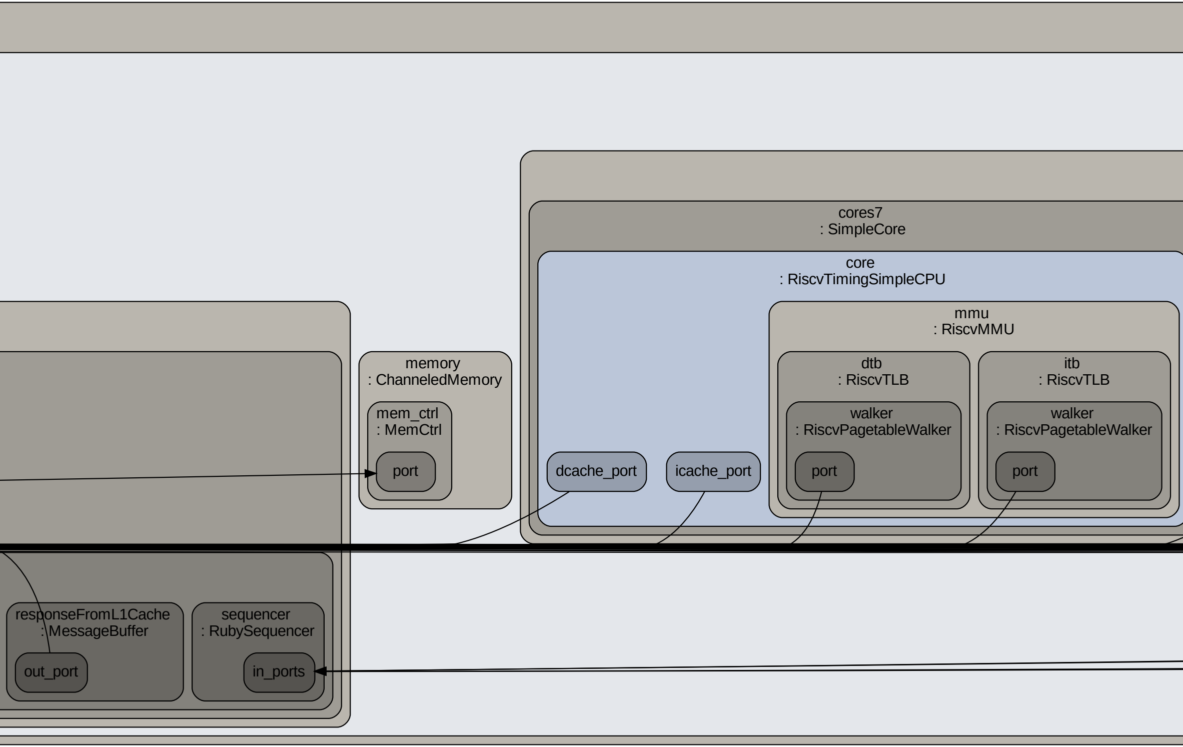

cache_hierarchy
: MESITwoLevelCacheHierarchyWithNetworks
ruby_system
: RubySystem
network
: SimplePt2Pt
out_port
in_port
responseFromL1Cache
: MessageBuffer
out_port
sequencer
: RubySequencer
in_ports
l1_controllers4
: L1Cache
requestToL1Cache
: MessageBuffer
in_port
responseToL1Cache
: MessageBuffer
in_port
requestFromL1Cache
: MessageBuffer
out_port
unblockFromL1Cache
: MessageBuffer
out_port

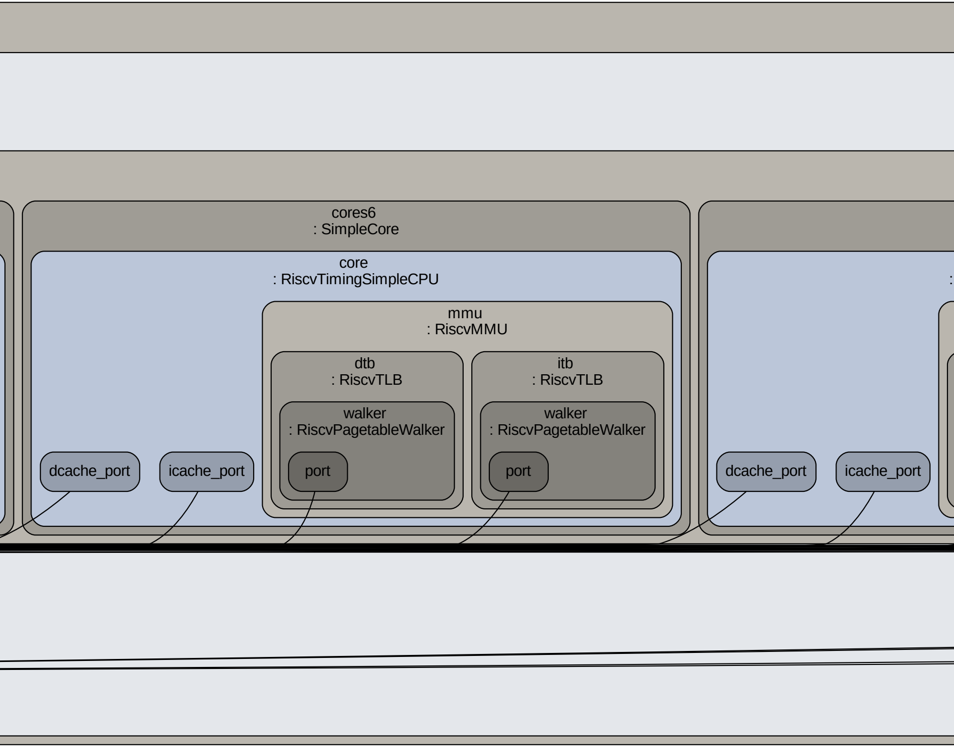

l1_controllers3
: L1Cache
responseFromL1Cache
: MessageBuffer
out_port
sequencer
: RubySequencer
in_ports
requestToL1Cache
: MessageBuffer
in_port
responseToL1Cache
: MessageBuffer
in_port
requestFromL1Cache
: MessageBuffer
out_port
unblockFromL1Cache
: MessageBuffer
out_port

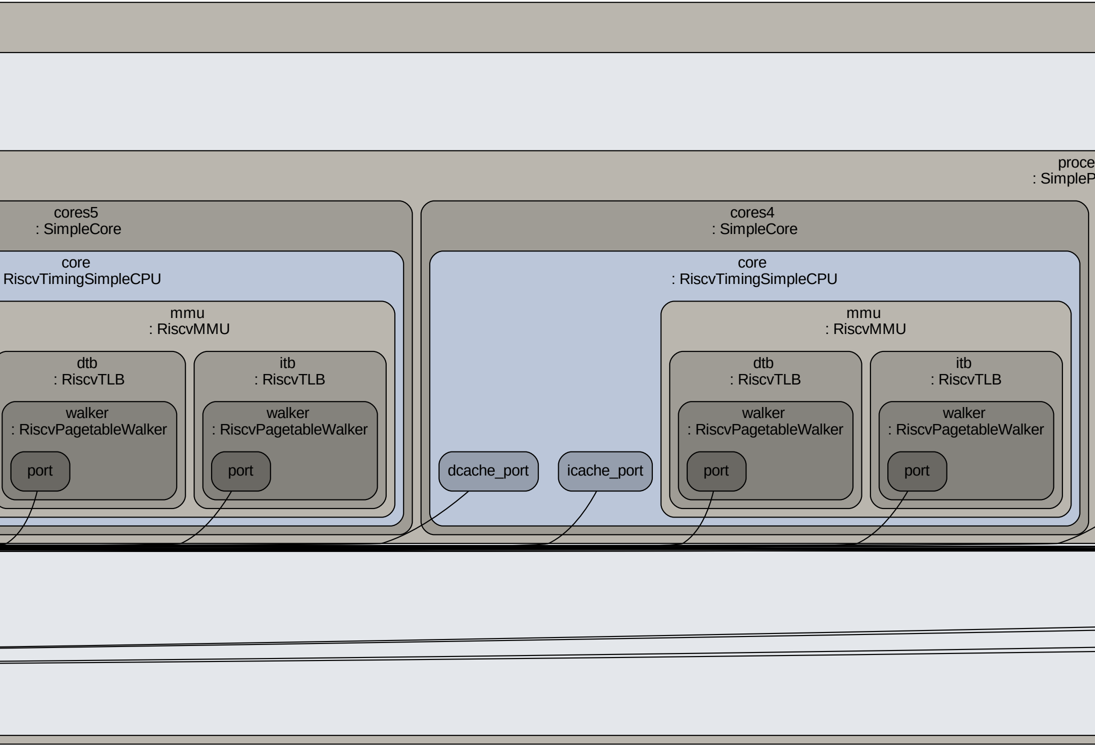

l1_controllers2
: L1Cache
responseFromL1Cache
: MessageBuffer
out_port
sequencer
: RubySequencer
in_ports
requestToL1Cache
: MessageBuffer
in_port
responseToL1Cache
: MessageBuffer
in_port
requestFromL1Cache
: MessageBuffer
out_port
unblockFromL1Cache
: MessageBuffer
out_port

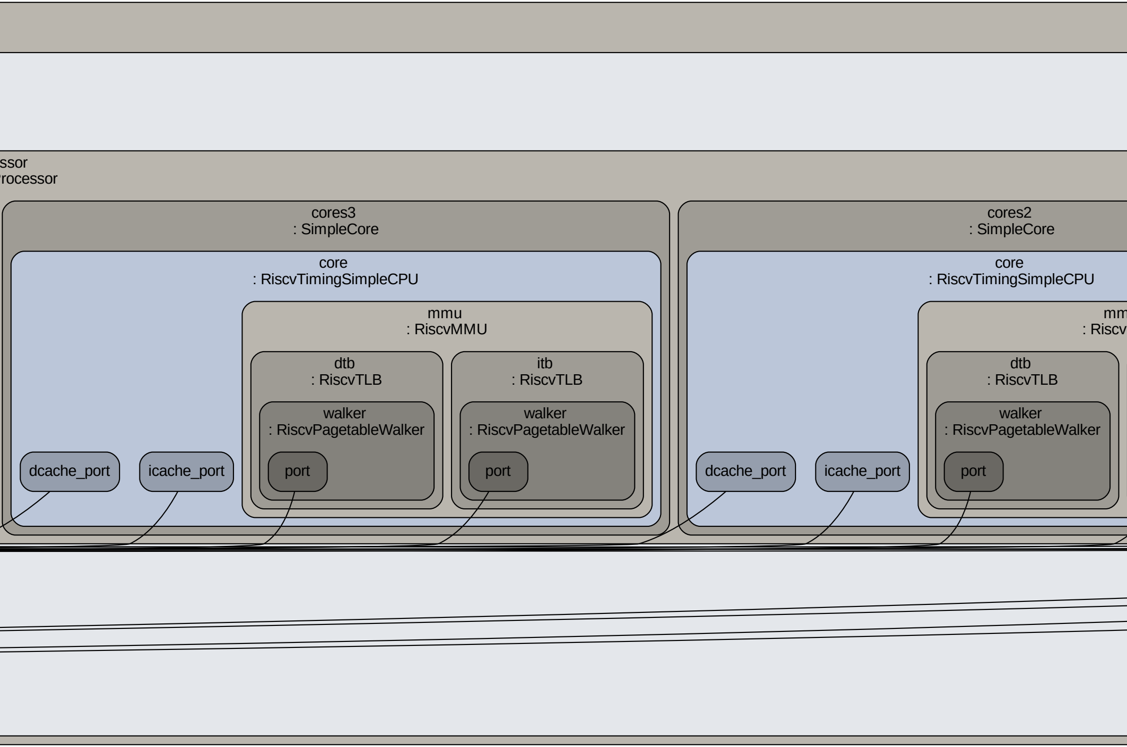

root
: Root
board
: SimpleBoard
l1_controllers1
: L1Cache
responseFromL1Cache
: MessageBuffer
out_port
sequencer
: RubySequencer
in_ports
requestToL1Cache
: MessageBuffer
in_port
responseToL1Cache
: MessageBuffer
in_port
requestFromL1Cache
: MessageBuffer
out_port
unblockFromL1Cache
: MessageBuffer
out_port

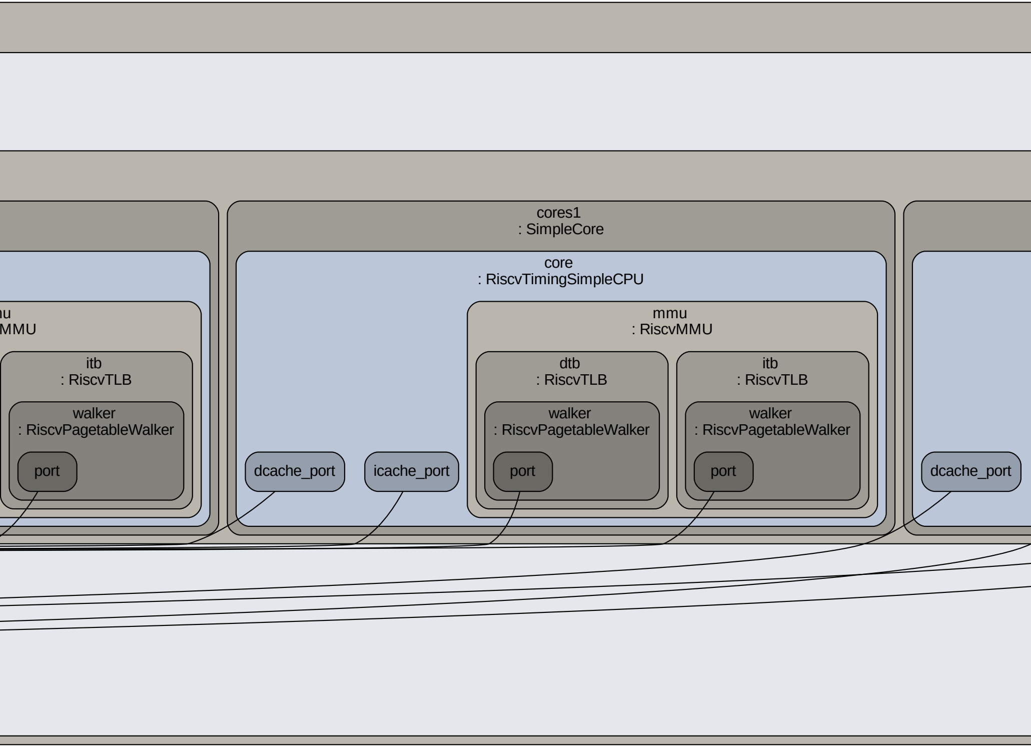

responseFromL1Cache
: MessageBuffer
out_port
sequencer
: RubySequencer
in_ports
l1_controllers0
: L1Cache
requestToL1Cache
: MessageBuffer
in_port
responseToL1Cache
: MessageBuffer
in_port
requestFromL1Cache
: MessageBuffer
out_port
unblockFromL1Cache
: MessageBuffer
out_port


cores7
: SimpleCore
core
: RiscvTimingSimpleCPU
mmu
: RiscvMMU
dtb
: RiscvTLB
itb
: RiscvTLB
walker
: RiscvPagetableWalker
walker
: RiscvPagetableWalker
port
port
dcache_port
icache_port
memory
: ChanneledMemory
mem_ctrl
: MemCtrl
port
responseFromL1Cache
: MessageBuffer
out_port
sequencer
: RubySequencer
in_ports

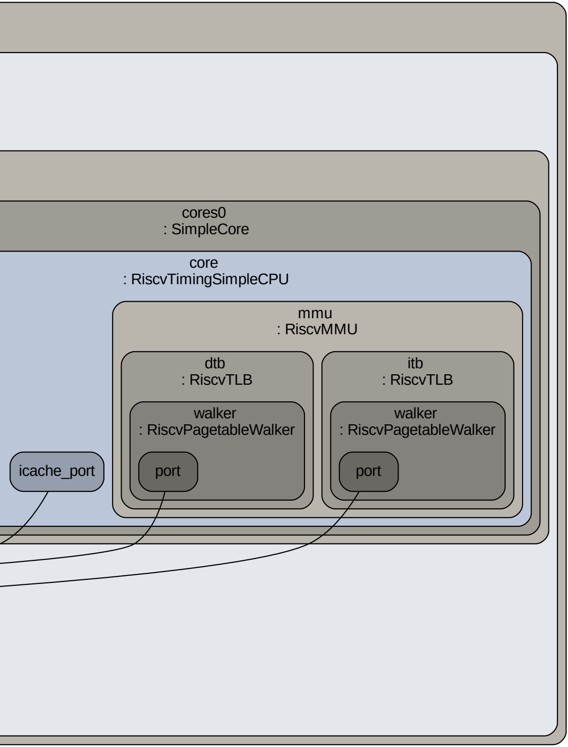

cores6
: SimpleCore
core
: RiscvTimingSimpleCPU
mmu
: RiscvMMU
dtb
: RiscvTLB
itb
: RiscvTLB
walker
: RiscvPagetableWalker
walker
: RiscvPagetableWalker
dcache_port
icache_port
port
port
dcache_port
icache_port


proce
: SimpleP
cores5
: SimpleCore
core
RiscvTimingSimpleCPU
mmu
: RiscvMMU
dtb
: RiscvTLB
walker
: RiscvPagetableWalker
port
itb
: RiscvTLB
walker
: RiscvPagetableWalker
port
cores4
: SimpleCore
core
: RiscvTimingSimpleCPU
dcache_port
icache_port
mmu
: RiscvMMU
dtb
: RiscvTLB
walker
: RiscvPagetableWalker
port
itb
: RiscvTLB
walker
: RiscvPagetableWalker
port


ssor
rocessor
cores3
: SimpleCore
core
: RiscvTimingSimpleCPU
mmu
: RiscvMMU
dtb
: RiscvTLB
walker
: RiscvPagetableWalker
port
itb
: RiscvTLB
walker
: RiscvPagetableWalker
port
dcache_port
icache_port
cores2
: SimpleCore
core
: RiscvTimingSimpleCPU
mm
: Riscv
dtb
: RiscvTLB
walker
: RiscvPagetableWalker
port
dcache_port
icache_port



nu
MMU
itb
: RiscvTLB
walker
: RiscvPagetableWalker
port
cores1
: SimpleCore
core
: RiscvTimingSimpleCPU
dcache_port
icache_port
mmu
: RiscvMMU
dtb
: RiscvTLB
walker
: RiscvPagetableWalker
port
itb
: RiscvTLB
walker
: RiscvPagetableWalker
port
dcache_port


cores0
: SimpleCore
core
: RiscvTimingSimpleCPU
mmu
: RiscvMMU
dtb
: RiscvTLB
itb
: RiscvTLB
walker
: RiscvPagetableWalker
walker
: RiscvPagetableWalker
icache_port
port
port